

NUCLEUS: Prompt Distribution Fidelity Determines Interactive Whole-Body PET/CT Lesion Segmentation Performance

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Abstract. We report a systematic investigation of interactive lesion segmentation for the autoPET V challenge, which evaluates algorithms on whole-body FDG and PSMA PET/CT with simulated user scribbles. Our central finding is methodological: the distribution of training prompts, not network architecture, dominated test performance. A four-channel ResEncL nnU-Net trained with dense oracle prompts reached 0.799 Dice on a patient-level held-out split but collapsed to 0.440 Dice on the challenge platform. Reading the organizers’ scribble simulator revealed the mechanism. Test-time foreground prompts cover only five randomly chosen lesion components, and background prompts are drawn from a two-voxel ring adjacent to true lesions; a model trained to suppress tissue near background prompts therefore deleted true disease. Regenerating training prompts with the organizers’ own simulation code, and forcing empty prompts on a quarter of cases, restored harness-condition performance from 0.569 to 0.752 Dice and recovered autonomous operation from 0.000 to 0.556 Dice. A fusion of the retrained model with the organizer baseline improved the hardest preliminary case above the baseline but lost ground on small PSMA lesions, and an ablation attributed that loss to boundary overshoot rather than component filtering. Stratifying detection by lesion CT density located the residual deficit in lesions below 100 HU. We additionally document a 29% patient-level leak in the default cross-validation split and nine label errors in the public cohort. All model comparisons were made against decision gates registered before the results were seen; the configuration submitted to the final phase is the organizer baseline, which none of our variants beat on preliminary mean position.

Keywords: autoPET challenge · Interactive segmentation · PET/CT · Prompt simulation · Data leakage.

1 Introduction

Interactive segmentation couples an automated model with sparse human guidance. The autoPET V challenge simulates this loop: algorithms receive a CT volume, a PET volume, and a set of foreground and background clicks, and are scored on voxel Dice and lesion-level F1. We entered the challenge with a

prompt-aware nnU-Net variant and, after a sequence of platform failures and score collapses, arrived at a conclusion we consider broadly useful. A prompt-aware model is only as good as the fidelity of its training-time prompt simulator to the test-time one. This paper quantifies that claim and adds data-quality observations, on split leakage and label errors, that are relevant to anyone training on the public autoPET cohort.

2 Data and Quality Audit

2.1 Cohort and label errors

We used the merged autoPET FDG and PSMA training release of 1,611 studies. Physician review of screening candidates identified nine studies with label errors, six of which showed disseminated osseous or nodal disease with empty or grossly incomplete ground truth. These six, together with four FDG studies whose label and image disagreed in axis orientation, were excluded, leaving 1,601 training studies.

2.2 Patient-level splits

The cohort contains longitudinal repeats of up to four studies per patient. The default case-level five-fold split of nnU-Net placed at least one study of 90 of 311 validation patients (29%) in the corresponding training folds. We rebuilt fold 0 at the patient level: 1,271 training and 330 validation studies (1,018 and 255 patients, zero overlap). All held-out numbers below use this split.

3 Methods

3.1 Prompt encoding and network

Prompts were encoded as two additional input channels using a clipped Euclidean distance transform, $\max(0, 1 - d/20)$ with d in voxels, around scribble voxels (foreground in channel 2, background in channel 3). The network was the nnU-Net ResEncL preset with four input channels, patch size $160 \times 224 \times 192$, batch Dice loss, and 250 epochs per fold [1, 2].

3.2 Prompt simulation: oracle versus harness-faithful

In the first configuration (v2, oracle prompts), foreground scribbles were generated on every ground-truth lesion, and background scribbles were sampled from real false-positive regions of a baseline model on the training set. In the second configuration (v3, harness-faithful prompts), we imported the organizers’ scribble simulator directly: foreground scribbles on five randomly selected ground-truth components using the centerline strategy, and background scribbles on five random components of the two-voxel dilation ring surrounding lesions. In addition, 25% of lesion-positive cases, and all lesion-free cases, received empty prompt channels, which forces the network to retain autonomous detection.

3.3 Post-processing and model fusion

Connected components smaller than five voxels were removed with 18-connectivity; the threshold came from a sweep on held predictions (+0.019 lesion F1 at a cost of 0.0002 Dice). We also evaluated a fusion of the prompt-insensitive organizer baseline with our strongest prompt-aware model: the fused mask is the union of the baseline prediction with those components of the prompt-aware prediction that overlap it. The intent is to keep the detection floor of the baseline while accepting boundary refinements only where both models agree a lesion exists.

3.4 Evaluation protocol

We used the official challenge metrics implementation throughout. For paired comparisons under harness conditions we built a 96-study internal set with prompts generated by the organizers’ simulator. These studies originate from the training pool, so absolute values are optimistic; the set was used strictly for paired comparisons against decision gates registered in advance, never for absolute claims. Platform numbers come from the five-case preliminary test set.

4 Results

4.1 Platform collapse of the oracle-prompted model

On the patient-level held-out split with its own dense prompts, v2 reached 0.799 Dice on fold 0, with five folds spanning 0.780 to 0.799 (mean 0.788). On the challenge platform the four-fold v2 ensemble scored 0.440 Dice and 0.598 F1, against 0.780 and 0.755 for the organizer baseline on the identical five cases. The failure signature was consistent across cases: true-positive volume halved and false negatives doubled on high-burden studies (0.844 to 0.236 Dice on the largest case), while false positives decreased. The model was suppressing true disease adjacent to ring-sampled background prompts and ignoring unprompted lesions.

4.2 Harness-faithful retraining

Table 1 summarizes paired results under harness-faithful prompts. Retraining with the organizers’ simulator recovered fold 0 from 0.569 to 0.743 Dice; a three-fold v3 ensemble with small-component filtering reached 0.752 Dice and 0.815 F1. Ensembling acted almost entirely through false-positive suppression (700 to 431 components), which localizes the residual deficit of v3 relative to the baseline in sparse-prompt delineation rather than detection. Against our registered gate, which required beating the filtered baseline on both metrics, the v3 ensemble cleared F1 by 0.007 and missed Dice by 0.013. Extending fold 0 to 1,000 epochs added 0.020 Dice (0.763, with 0.824 F1), and its fusion with the baseline reached 0.766 Dice and 0.812 F1, the first configuration to clear the gate on both columns; it was therefore evaluated on the platform.

Table 1. Paired comparison under harness-faithful prompts (96 internal studies of training-pool origin; paired use only). The filter removes components smaller than 5 voxels. Counts are lesion instances summed over studies.

Configuration	Dice	F1	TP	FP	FN
v2 ensemble (4 folds), filtered	0.5686	0.7281	807	367	731
Organizer baseline (native prompts)	0.7650	0.7981	1141	567	397
Organizer baseline, filtered (gate)	0.7641	0.8081	1115	467	423
v3 fold 0	0.7428	0.7471	1138	700	400
v3 ensemble (3 folds), filtered	0.7515	0.8150	1092	431	446
v3 fold 0, 1,000 epochs	0.7632	0.8236	1092	393	446
Fusion of baseline and 1,000-epoch model	0.7656	0.8115	1103	405	435

4.3 Autonomous operation

With empty prompt channels, v2 produced no output at all: 0.000 Dice across 96 studies. Dense oracle prompting had taught pure prompt following. The v3 model, trained with 25% empty prompts, scored 0.556 Dice and 0.541 F1 without any guidance, recovering genuine detection while keeping prompt responsiveness (0.743 with prompts).

4.4 Preliminary test set

The fused container completed all five preliminary cases and scored 0.7558 Dice and 0.7625 F1, against 0.7800 and 0.7554 for the baseline. The fusion improved the highest-burden FDG case above the baseline (0.866 versus 0.844) but lost Dice on both PSMA cases (0.528 versus 0.605, and 0.719 versus 0.752). Because the fusion can only add voxels to the baseline mask, the PSMA losses could stem either from the component filter or from the added voxels themselves. A second platform run with the filter threshold lowered from five voxels to three left Dice essentially unchanged (0.7553) and reduced F1 (0.7589), which attributes the PSMA deficit to boundary overshoot of the prompt-aware arm on small lesions rather than to component filtering. The trade is tracer dependent: fusion helps bulky FDG disease and penalizes small PSMA lesions.

4.5 Detection stratified by lesion CT density

Motivated by a clinical impression during case review that sclerotic, tracer-avid lesions were being missed, we stratified all 1,725 ground-truth lesions of the internal harness set by mean CT attenuation. The impression inverted under measurement. Per-lesion detection sensitivity was highest for sclerotic lesions above 300 HU (87.3% baseline, 88.0% fused) and lowest for lytic or soft-tissue-density lesions below 100 HU (71.3% and 72.8%), with mixed-density lesions between (78.0% and 79.1%). The fused configuration matched or exceeded the baseline in every density bin. The residual detection deficit of all configurations therefore lies in lesions with low CT contrast; the cross-bin gradient, not the absolute rates, is the finding, given the training-pool origin of the set.

4.6 Negative results

We report measured dead ends for completeness. Test-time mirroring augmentation changed Dice by $+0.0005$ and reduced F1 by 0.005 at eight times the inference cost. A probability-threshold sweep was flat, 0.791 to 0.793 Dice across thresholds 0.2 to 0.7, indicating well-calibrated confidences. Organ-prior false-positive suppression failed because 103 of 106 false-positive components in our worst case had no overlap with any TotalSegmentator organ, with attenuation values indicating predominantly osseous locations. A plain union of prompt-aware and prompt-free predictions scored 0.7623 and 0.7932, below the gate.

5 Discussion

Three observations generalize beyond our entry. First, prompt distribution fidelity is a first-order design variable. A swing of 0.36 Dice separated the same architecture trained under oracle versus harness-faithful prompting, which exceeds every architecture and post-processing effect we measured by an order of magnitude. The background-prompt semantics were decisive: ring-adjacent background scribbles are corrections of boundary overshoot, not vetoes of regional disease, and a model must be trained under exactly that semantics. Second, empty-prompt forcing is cheap insurance. A 25% empty-prompt fraction converted a model that was inert without guidance into one retaining 0.556 autonomous Dice, at no measured cost to prompted performance. Third, public-cohort hygiene matters. A 29% patient-level validation leak and nine label errors, six with disseminated disease annotated as lesion-free, were each capable of misleading model selection by several Dice points. We recommend patient-level splitting and a targeted audit of empty-ground-truth studies to all users of this cohort.

The preliminary results also illustrate the limits of late fusion. Keeping the baseline’s detection floor and accepting prompt-aware refinements only where both models agree improved the hardest case, yet the refinements themselves overshoot on small PSMA lesions, and no filter threshold repairs that. Closing the gap appears to require training-time exposure to the deployment prompt distribution at every scale of lesion, not post hoc mask arithmetic.

6 Conclusion

Our submitted configuration is the organizer nnU-Net baseline, retained after prompt-aware variants failed a registered improvement gate on preliminary mean position. The contribution of this work is a controlled characterization of why prompt-aware training fails or succeeds in this setting, together with the accompanying data-quality findings and a density-stratified localization of the remaining detection deficit.

References

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